PIERLUIGI RINALDI
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RESEARCH INTERESTS & TECHNICAL SKILLS

Galaxy formation and evolution at intermediate and high redshifts; stellar mass assembly; star formation activity and its evolution over cosmic time; galaxy morphology; infrared-selected galaxies; Epoch of Reionization and source of Cosmic Reionization; Emission line galaxies; Extremely Red Objects; multi-wavelength photometric and spectroscopic follow-up; active galactic nuclei and supermassive black hole growth; galaxy–SMBH co-evolution across cosmic time; large-scale structures; stellar population synthesis models and spectral energy distribution techniques; instrumentation and software development for JWST.

- **Coding:** Python, LaTeX, C++, bash.
- **Data Reduction:** Reduction of JWST data (MIRI, NIRCam, and NIRISS imaging; NIRCam/WFSS; NIRSpec).
- **Data Analysis:** UV/Optical/NIR (HST) and NIR/MIR (JWST) imaging and spectroscopy; Spectral Energy Distribution (SED) fitting; photometry; multi-wavelength catalogue creation; spectral line analysis, morpholgical analsys, environmental analysis, mock data and catalogs.
- **Software:** JWST pipeline (MIRI, NIRCam, NIRISS, NIRSpec), Bagpipes, Prospector, LePHARE, EAZY, Grizli, MSAEXP, MIRISim (to simulate MIRI data), MIRAGE (to simulate JWST/NIRCam, NIRISS, FGS), SourceExtractor, GALFIT, forcepho.

ACADEMIC POSITION (after PhD)

- **Postdoctoral Research Fellow** at Space Telescope Science Institute (STScI) (August 2025 - present)
  (Mentor: Stacey Alberts)

- **Postdoctoral Research Associate I** at Steward Observatory, University of Arizona (August 2024 - August 2025)
  (Mentors: George Rieke & Stacey Alberts)

EDUCATION

- **Ph. D. in Astrophysics** (December 10th, 2024)
  Kapteyn Astronomical Institute - University of Groningen
  Title: Unveiling the early Universe with JWST
  Adivsors: Karina I. Caputi & Filippo Fraternali

- **M.Sc. in Astrophysics & Cosmology** (March 13th, 2020)
  Alma Mater Studiorum - University of Bologna
  Title: Simulation of high-redshift galaxy spectra in the Euclid Deep Fields
  Advisors: Andrea Cimatti & Michele E. M. Moresco

- **B.Sc. in Physics** (September 20th, 2017)
  University of Bari Aldo Moro
  Title: The condensation - from clouds to protoplanetary systems
  Advisor: Savino Longo

SELECTED CONFERENCES AND SEMINARS (CHRONOLOGICAL ORDER)

- **Invited Talk** / Seminar — Space Telescope Science Institute (Baltimore, US) [2025]
- **Poster** / Conference — The Inaugural Cosmic Frontier Center Conference (Austin, US) [2025]
- **Talk** / Conference — First Galaxies (Oxford, UK) [2025]
- **Selected Talk**/ Consortium Meeting — The MIRI EU meeting (Madrid, Spain) [2025]
- **Selected Talk** / Consortium Meeting — The NIRCam meeting (Biosphere 2, Tucson, US) [2025]
- **Talk**/ Invited Seminar — Kapteyn Astronomical Institute (Groningen, NL) [2024]
- **Selected Talk** / Consortium Meeting — The MIRI US and EU meeting (Biosphere 2, Tucson, US) [2024]
- **Invited Talk** / Seminar — Scuola Internazionale Superiore di Studi Avanzati (Trieste, IT) [2024]

- **Invited Talk** / Seminar — Space Telescope Science Institute (Baltimore, US) [2024]
- **Talk** / Conference — Netherlands Research School for Astronomy (Leiden, NL) [2023]
- **Invited talk** / Consortium Meeting — MIRI European Consortium meeting, plenary talk on *Highlights of the MIRIM science in the Hubble eXtreme Deep Field* (Dublin, IRE) [2023]
- **Talk** / Conference — The James Webb Space Telescope turns one: the birth and growth of galaxies (Sexten Center for Astrophysics, Sesto, IT) [2023]
- **Invited Talk** / Conference — Shedding new light on the first billion years of the Universe (Marseille, FR) [2023]
- **Invited Talk** / Seminar — INAF/OAS Padova (Padova, IT) [2022]
- **Invited Talk** / Seminar — Astrophysics Talks at INAF/OAS Bologna (Bologna, IT) [2022]
- **Poster** / Conference — European Astronomical Society (Valencia, ES) [2022]
- **Talk** / Conference — Netherlands Research School for Astronomy (online) [2021]

GRANTS & AWARDS

- 222K USD for JWST GO 5578 (JWST, cycle 3) [2024]
- Royal Astronomical Society Group Achievement Award for JWST-MIRI Team [2024]

SUCCESSFUL TELESCOPE PROPOSALS

- **Co-Principal Investigator** of "The MIRI deep imaging survey of the lensing clusters Abell2744 and MACS0416" (JWST cycle 3; PID: 5578) [75h]
- **Co-Investigator of the following proposal:**
  PID: 9433 (JWST, DDT), PID: 7935 (JWST, cycle 4); PID: 6511 (JWST, cycle 4), PID: 7605 (JWST, cycle 4), PID: 8018 (JWST cycle 4), PID: 8051 (JWST, cycle 4), PID: 8544 (JWST, cycle 4), PID: 2926 (JWST, cycle 2), PID: 3432 (JWST, cycle 2), PID: 3538 (JWST, cycle 2), PID: 1283 (JWST, Cycle 1), ALMA Cycle 12 (2025.1.01067.S), ALMA Cycle 12 (2025.1.00147.S) [300h+ in total]

COLLABORATION MEMBERSHIP

- **Co-Principial Investigator** with Edoardo Iani of the MIRIC team and **Leader of the MIRI data reduction** (GO 5578) [2024 - present]
- **Key member and (junior) co-leader for MIRI data reduction and co-leader of the NIRCam data reduction** of the MIDIS team [2021 - present]
- **Core Member** of the MIRI European and US consortium [2021-present]
- **Key member** of the SMILES team [2024 - present]
- **Core member** of the OASIS team [2024 - present]
- **Core member** of the JADES team [2024 - present]
- **Member** of the Nancy Grace Roman Telescope Working Groups (software, simulations, synergies, and calibration) [2025 - present]
- **Member** of extragalacitc working group (galaxy evolution and AGNs) of the PRIMA Probe far-Infrared Mission for Astrophysics [2025-present]
- **Fromer member** of the Euclid consortium [2019-2021]

SUPERVISION, MENTORING AND TEACHING

- (Co-Supervisor) Bachelor Student — Nadav Peleg-Brochestein at Rijksuniversiteit Groningen (2024)
- (Mentor) PhD Student — Rafael Navarro-Carrera at Rijksuniversiteit Groningen (2022-2024) [**with two publications — see publication list]**
- (Teaching Assistant) — Astrophysical Hydrodynamics, Bachelor course, Science and Engineering, Rijksuniversiteit Groningen (2021-2024)

- (Teaching Assistant) — Physics, First cycle degree programme (L) in Biological Sciences (cod. 8012), Biological Sciences, Alma Mater Studiorum - University of Bologna (2019-2020)

PROFESSIONAL SERVICE TO THE ASTRONOMICAL COMMUNITY

- **Reviwer** for scientific journals
- **Panel Support Scientists** for JWST Cy5 [2025-present]

PRESS RELEASES

- Press release in preparation with STScI/NASA on a candidate descendant of a Little Red Dot at Cosmic Noon [December 2025]
- Press release in preparation with NOVA on the discovery of Little Red Dots, revealing complex UV morphologies rather than appearing as mere dots [October 2025].
- MIRI instrument on JWST detects H-alpha emission during the Epoch of Reionization for the first time [2023]
- The James Webb telescope detects much more light than expected coming from the primaeval universe [2023]
- Astronomers see the infrared switch on in young universe [2022]
- Early Universe bristled with starburst galaxies [2022]

OUTREACH

- Talk at Associazione Astronomiamo, Milan (IT) — Beyond Hubble: Unfolding the Early Universe with JWST [2024]
- Local Organizing Committee— 6th NOVA evaluation, Groningen [2024].
- Talk at Kapteyn Science Day [2022]
- Local Organizing Committee — JWST First Data Release - Public Presentation in Groningen, DOT Planetarium [2022]
- Member of the Astronomy on Tap in Groningen [2022-2023]

LIST OF PUBLICATIONS

Summary:

✤ Submitted and Published papers: 72 (ApJ, A&A, MNRAS, Nature, etc.). Citations:**1000+**. H-index = **23** (source: NASA ADS).

✤ Total of 1st, 2nd, and 3rd author submitted and/or published papers: **7**, **5**, and **2**. Citations: **300+** (**65% as first-author**).

✤ Publication record: **NASA ADS/Pierluigi Rinaldi**

In the process of submission at the moment of writing this document:

✤ **P. Rinaldi**, S. Alberts et al. 2025 — "Bron in Fire: JWST Reveals a Star-Forming Proto-Cluster at Cosmic Noon"

First-, second-, and third-author papers (already submitted/accepted):

✤ **P. Rinaldi**, G. H. Rieke et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250717738R/abstract

✤ E. Iani, **P. Rinaldi** et al. 2025 — https://ui.adsabs.harvard.edu/abs/2024arXiv240618207I/abstract

✤ **P. Rinaldi**, P. G. Pérez-González et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250401852R/abstract

✤ **P. Rinaldi**, N. Bonaventura et al. 2025 — https://ui.adsabs.harvard.edu/abs/2024arXiv241114383R/abstract

✤ R. Navarro-Carrera, **P. Rinaldi** et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024arXiv241023249N/abstract

✤ **P. Rinaldi**, R. Navarro-Carrera et al. 2025 — https://ui.adsabs.harvard.edu/abs/2024arXiv240613554R/abstract

✤ P. G. Pérez-González, **P. Rinaldi** et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024arXiv240216942P/abstract

✤ K. I. Caputi, **P. Rinaldi** et al. 2024 — https://ui.adsabs.harvard.edu/abs/2023arXiv231112691C/abstract

✤ **P. Rinaldi**, K. I. Caputi et al. 2024 — https://ui.adsabs.harvard.edu/abs/2023arXiv230915671R/abstract

✤ R. Navarro-Carrera, **P. Rinaldi** et al. 2024 — https://ui.adsabs.harvard.edu/abs/2023arXiv230516141N/abstract

✤ E. Iani, K. I. Caputi, **P. Rinaldi** et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024ApJ...963...97I/abstract

✤ **P. Rinaldi**, K. I. Caputi et al. 2023 — https://ui.adsabs.harvard.edu/abs/2023ApJ...952..143R/abstract

✤ E. Iani, K. I. Caputi, **P. Rinaldi** et al. 2022 — https://ui.adsabs.harvard.edu/abs/2022ApJ...940L..24I/abstract

✤ **P. Rinaldi**, K. I. Caputi et al. 2022 — https://ui.adsabs.harvard.edu/abs/2022ApJ...930..128R/abstract

Other publications I co-authored:

✤ Shivaei, I. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250901795S

✤ Ormerod, K. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025MNRAS.542.1136O

✤ D'Eugenio, F. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025MNRAS.542..960D

✤ Zhu, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250900153Z

✤ Annunziatella, M. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250816951A

✤ Zhu, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250812599Z

✤ Ji, Z. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...988L..69J

✤ Witstok, J. et al. 2025— https://ui.adsabs.harvard.edu/abs/2025arXiv250722888W

✤ Wu, Z. — https://ui.adsabs.harvard.edu/abs/2025arXiv250722858W

✤ Scholtz, J. — https://ui.adsabs.harvard.edu/abs/2025arXiv250717738R

✤ Prieto-Jimenez, C. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250706793P
✤ Isobe, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025MNRAS.541L..71I
✤ Puskas, D. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025MNRAS.540.2146P
✤ Laseter, L. H. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...988...73L
✤ Rieke, G. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...988...17R
✤ Costantin, L. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025A&A...699A.360C
✤ Geris, S. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250622147G
✤ D'Eugenio, F. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250614870D
✤ Helton, J. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250602099H
✤ Zhu, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...986..162Z
✤ Zhu, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...986...18Z
✤ Lin, X. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250502896L
✤ Zhang, J. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250502895Z
✤ Baker, W. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025A&A...697A..90B
✤ Baker, W. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025MNRAS.539..557B
✤ Sun, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250414682S
✤ Lin, X. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250408028L
✤ Juodzbalis, I. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250403551J
✤ Sun, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...983..165S
✤ Ostlin, O. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025A&A...696A..57O
✤ Danhaive, L. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250321863D
✤ Perez-Gonzalez, P. G. et al. 2025— https://ui.adsabs.harvard.edu/abs/2025arXiv250315594P
✤ Scholtz, J. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250310751S
✤ Witstok, J. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025Natur.639..897W
✤ Alvarez-Marquez, J. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025A&A...695A.250A
✤ Cooper, R. A. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250218310C
✤ Hainline, K. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...979..138H
✤ Bik, A. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025A&A...694C...1B
✤ Gillman, S. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250111491G
✤ Moriya, T. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250108969M
✤ Coulter, D. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250105513C
✤ Whitler, L. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025arXiv250100984W
✤ Sun, Y. et al. 2025 — https://ui.adsabs.harvard.edu/abs/2025ApJ...978...98S
✤ Simmonds, C. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024MNRAS.535.2998S
✤ Crespo Gomez, A. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024A&A...691A.325C
✤ Bosman, S. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024NatAs...8.1054B
✤ Navarro-Carrera, R. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024arXiv240714201N
✤ Boogaard, L. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024ApJ...969...27B
✤ Kokorev, V. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024ApJ...968...38K
✤ Alvarez-Marquez, J. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024A&A...686A..85A
✤ Bik, A. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024A&A...686A...3B
✤ van Mierlo, S. et al. 2024 — https://ui.adsabs.harvard.edu/abs/2024ApJ...966...12V
✤ Yang, G. et al. 2023 — https://ui.adsabs.harvard.edu/abs/2023ApJ...956L..12Y
✤ Perez-Gonzalez, P. G. et al. 2023 — https://ui.adsabs.harvard.edu/abs/2023ApJ...951L...1P
✤ Yang, G. et al. 2023 — https://ui.adsabs.harvard.edu/abs/2023ApJ...950L...5Y
✤ Colina, L. et al. 2023 — https://ui.adsabs.harvard.edu/abs/2023A&A...673L...6C
✤ Kokorev, V. et al. 2023 — https://ui.adsabs.harvard.edu/abs/2023ApJ...945L..25K
✤ Iani, E. et al. 2023 — https://ui.adsabs.harvard.edu/abs/2023MNRAS.518.5018I